

25/8/26

CS340

Lecture 12

Uniqueness of Minimum
state DFA

Minimum Size DFA

Let L be a regular language. From the construction in proof of Myhill-Nerode theorem, we know that L has a DFA M_L with number of states equal to the index of R_L . Further, any other DFA M recognizing L has number of states $\geq$ index of R_L . It follows that M_L is ~~the~~ a minimum ~~states~~ automaton to recognize L .

What is more, is that there is only one minimum size automaton (upto isomorphism) to recognize L .

Defn (Isomorphism of DFAs)

Let $M = (Q, \Sigma, \delta_M, q_0, F_M)$ and

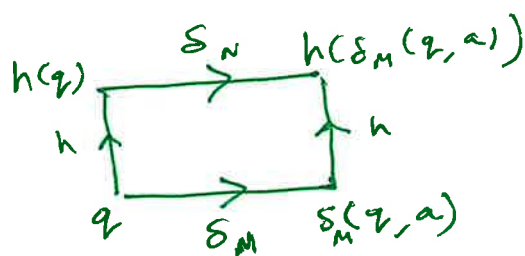
$N = (R, \Sigma, \delta_N, r_0, F_N)$ be two DFAs.

$h: Q \rightarrow R$ is an isomorphism from M to N iff h is a bijection and for all $q \in Q, a \in \Sigma$

$$(i) \quad h(\delta_M(q, a)) = \delta_N(h(q), a)$$

That is, for each $a \in \Sigma$,

Diagram



Commutes,

$$(ii) \quad h(q_0) = r_0$$

$$(iii) \quad q \in F_M \iff h(q) \in F_N$$

Isomorphism h may be considered just renaming of states of M , which ~~preserves structure of automaton~~. In the above def, we just changed name of state q of M to $h(q)$ without making any ^{other} change to its transition f , initial or final states. The resulting automaton is N .

Theorem Let M be a minimum state automaton to recognize L . M is isomorphic to M_L .

Proof:

$$M = (Q, \Sigma, q_0, \delta_M, F)$$

$|Q| =$ index of R_L .

It follows that all states in Q are reachable from q_0 (otherwise, we could ~~the~~ remove unreachable states from Q and recognize L by an automaton with lesser no. of states than is index of R_L , a contradiction)

$\Rightarrow$ ~~the~~ ~~the~~ R_M has exactly $|Q|$ equivalence classes. Equiv class of x is given as

$$[x] = \{ y \in \Sigma^* \mid \delta_M(q_0, y) = \delta_M(q_0, x) \}$$

As R_M refines R_L (equivalence classes of R_M refine equiv classes of R_L), ~~the~~ the equiv classes of R_M are the same as the ~~the~~ equiv classes of R_L . **In fact, $R_M = R_L$**

~~There is a bijection~~ ^{be} Each equiv class of R_M is associated with ~~the~~ ^a unique state ~~of~~ in Q . It follows that each equiv class of R_L is associated to a unique state in Q .

Define $h: Q_L \rightarrow Q$ stated above

that is $h([x]) = \delta_M(q_0, x)$

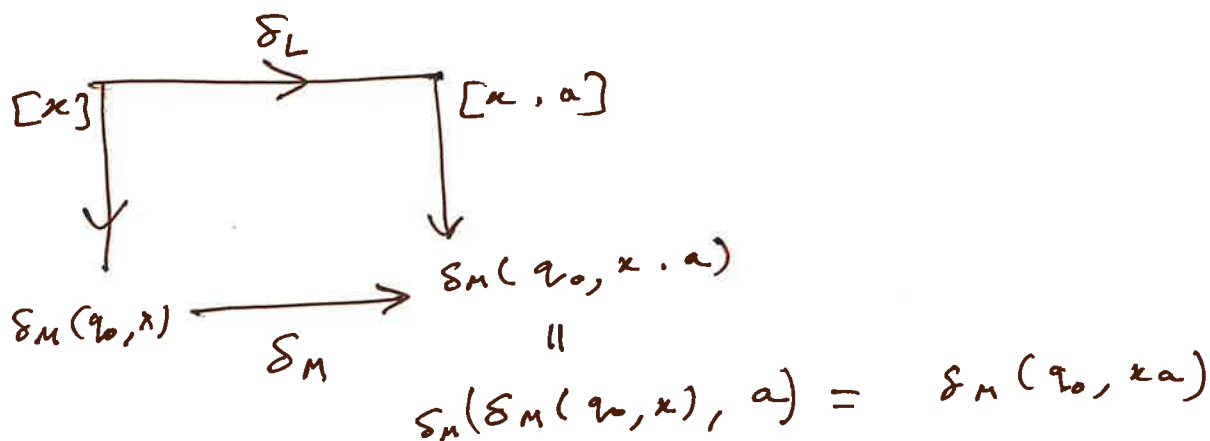
h is well defined,

$$[x] = [y] \Rightarrow x R_L y \Rightarrow x R_M y \quad (\because R_L = R_M)$$

$$\delta_M(q_0, x) = \delta_M(q_0, y) \Rightarrow h([x]) = h([y])$$

~~$\therefore \delta_M(q_0, x)$ has the class $[x]$ associated to it~~

We need to show



$q_0 = h([\epsilon])$ (by def of $\delta_M(q_0, \epsilon) = q_0$)

~~$[x] \in F_L \iff x \in L$~~

$$[x] \in F_L \iff x \in L \iff \delta_M(q_0, x) \in F$$

$$\iff h([x]) \in F.$$

□